POLITEKNIK MALAYSIA
TUN SYED NASIR

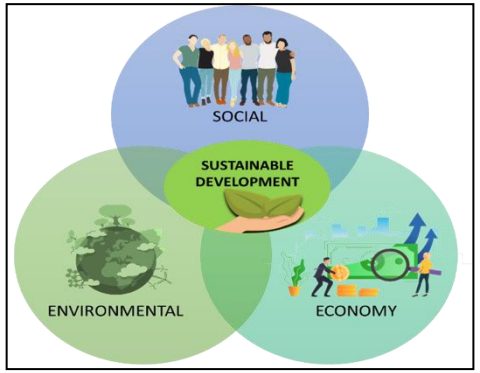


CHAPTER 5: CURRENT ISSUE RELATED TO ENVIRONMENT | Ts. MUHAMMAD AZRI IZANI BIN MOHAMAD HALIM

INTRODUCTION

Sustainable development defined as development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

- Sustainable development also involves balancing three main aspects:
- **Environmental protection:** Conserving natural resources and reducing pollution
- **Social equity:** Ensuring fairness, well-being, and access to basic needs for all people
- **Economic development:** Supporting growth while maintaining long-term resource availability



CURRENT ISSUES RELATED TO O&F INDUSTRY

- The oil and fat industry, especially the palm oil sector in Malaysia, faces several environmental challenges linked to production, processing, and land use.
- The palm oil industry plays a major role in global edible oil supply but also contributes significantly to environmental issues such as:
    1. Deforestation
    2. Climate change & global warming
    3. Ozone layer depletion
    4. Acid rain
    5. Carbon footprint
    6. Air & water pollution

DEFORESTATION*

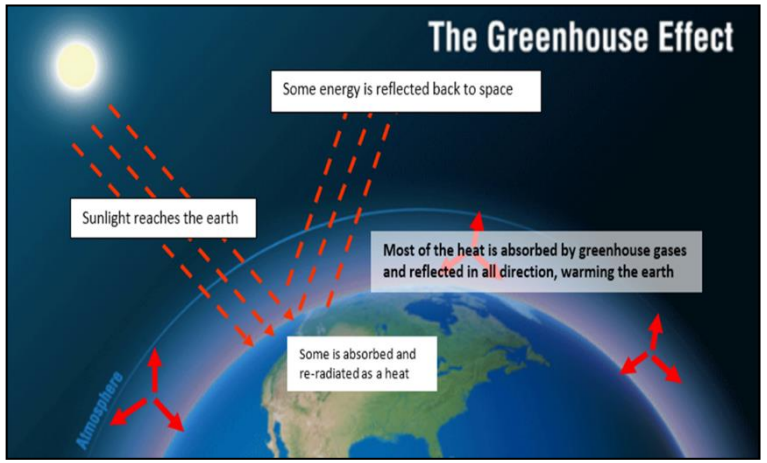
- Deforestation occurs when a forest or stand of trees is removed from land and transferred to a non-forest use, such as conversion of forest land to farmland, ranches, or urban usage.
- Palm oil production has become the major cause of deforestation in the world due to high global demand for palm oil.
- Impact of deforestation:



1. Disruption to the forest ecosystem and loss of species such as orang utan.
2. Increased greenhouse gas emissions which lead to climate change.
3. Decreases the carbon sink, and the decay of the resulting dead plant matter contribute to a new carbon source.

CLIMATE CHANGE & GLOBAL WARMING

- Climate change is defined as a change in the average conditions such as temperature and rainfall in a region over a long period of time.
- Global warming is defined as the increase in mean surface temperatures on earth as a result of human activities, primarily fossil fuel burning, which increases heat-trapping greenhouse gas levels in Earth's atmosphere.
- Palm oil production contributes to greenhouse gas (GHG) emissions through these activities:
    1. Land clearing
    2. Fertilizer use
    3. Palm oil mill effluent (POME) decomposition
    4. Open burning
    5. Fuel use (mills & transport)



CLIMATE CHANGE & GLOBAL WARMING

- Impacts of Climate Change and Global Warming:

1. **Sea Level Rise**
    - Global warming causes melting of glaciers and polar ice caps.
2. **Ocean Acidification**
    - Increased CO_2 absorption by oceans forms carbonic acid will educes ocean pH, making seawater more acidic.
3. **Extreme Weather Events (Droughts, Floods, Heatwaves)**
    - Droughts reduce water availability and agricultural productivity.
    - Floods destroy crops, soil quality, and infrastructure.
    - Extreme temperatures stress both crops and animal.
4. **Impact on Wildlife and Biodiversity**
    - Rising temperatures alter natural habitats and ecosystems.
    - Changes in rainfall and vegetation patterns force species migration.

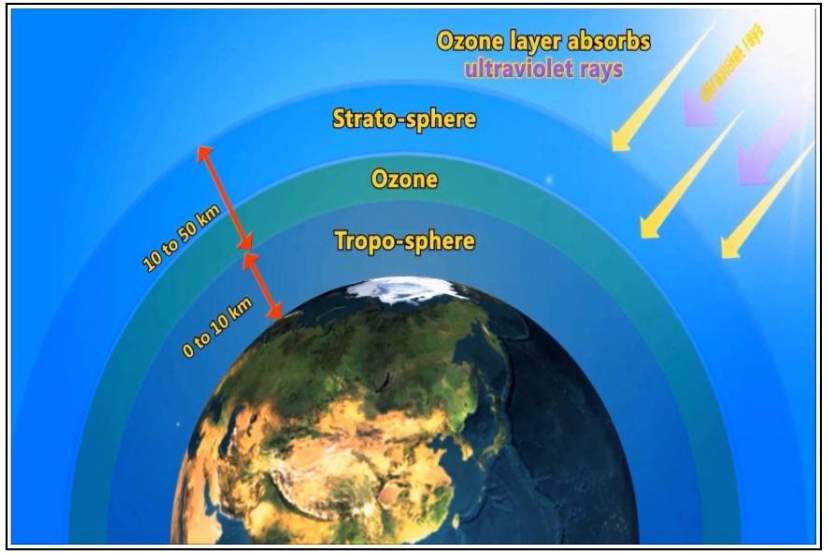
EXAMPLE 1

- State 5 activities from palm oil industry that contribute to GHG emission
- Explain 2 impact of climate change and global warming.

OZONE LAYER DEPLETION

- Ozone (O_3) is a highly reactive gas composed of three oxygen atoms.
- This layer plays a critical role in protecting life on Earth by absorbing a significant portion of the sun's harmful ultraviolet (UVB) radiation, thereby reducing harmful exposure to living organisms.
- Ozone depletion occurred due to ozone depleting substances (ODS) from human activity and industries is released to atmosphere.
- Impact of ozone depletion:

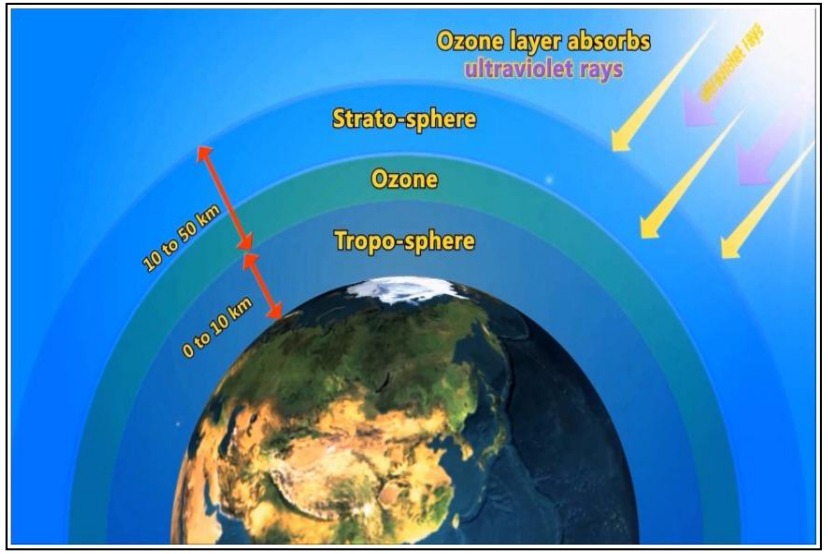
1. Reduced plant growth and productivity
2. Damage to photosynthesis process
3. Lower crop yield and agricultural output
4. Increased risk of skin cancer due to higher UV exposure
5. Eye damage such as cataracts



OZONE LAYER DEPLETION

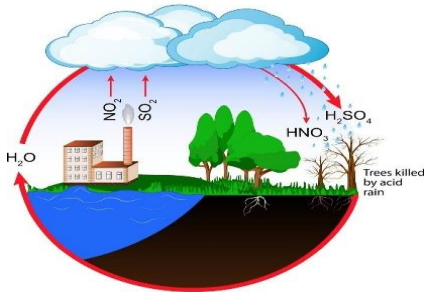
Ozone depletion formation:

1. Release of **ozone-depleting substances (ODS)** from human activities and industries
2. Common ODS include chlorofluorocarbons (CFCs), halons, and related chemicals
3. These substances rise to the stratosphere and break down ozone molecules
4. Leads to gradual thinning of the ozone layer



ACID RAIN

- Acid rain describes any form of precipitation which contains high levels of nitric and sulfuric acids that falls to the earth from the atmosphere in both dry and wet form.
- Normal rain is slightly acidic, with a pH of 5.6, while acid rain generally has a pH between 4.2 and 4.4.
- Formation of Acid Rain (Related to Palm Oil Industry):



1. In some palm oil mills, coal or natural gas is used as fuel in boilers to generate heat and electricity
2. Combustion of fossil fuels releases sulfur dioxide (SO_2) into the atmosphere
3. High-temperature combustion processes also produce nitrogen oxides (NO_x)
4. Additional SO_2 and NO_x are released from transportation activities and oil refining processes
5. In the atmosphere, SO_2 and NO_x undergo chemical reactions with water vapor, oxygen, and other atmospheric gases
6. These reactions form sulfuric acid (H_2SO_4) and nitric acid (HNO_3)
7. The acids dissolve in cloud moisture and are transported over long distances by wind
8. Eventually, they return to Earth's surface as acid rain

ACID RAIN

- Impact of acid rain:

1. Acid rain makes soil and water bodies more acidic and toxic, leading to increased aluminium release from soil, which can be transported into lakes and rivers and harm aquatic life.
2. It also weakens trees and plants by reducing soil fertility and nutrient availability, making them more vulnerable to cold stress, insect attacks, and diseases.
3. Sulfur dioxide (SO_2) and nitrogen oxides (NO_x) in the atmosphere can form fine sulfate and nitrate particles, which can be inhaled by humans and cause respiratory problems and breathing difficulties.
4. Acid rain also damages buildings, monuments, and infrastructure, especially those made of limestone and marble, causing surface erosion, material loss, and the gradual destruction of architectural details.
5. Acid rain leads to the acidification of lakes and rivers, resulting in lower pH levels that harm fish and other aquatic organisms and disrupt aquatic ecosystems.

EXAMPLE 2

- Explain formation of acid rain.

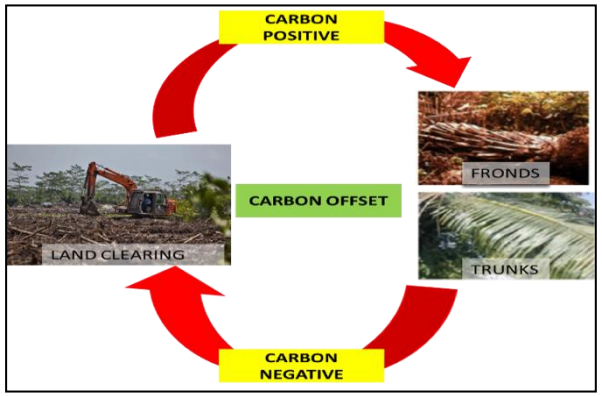
CARBON FOOTPRINT

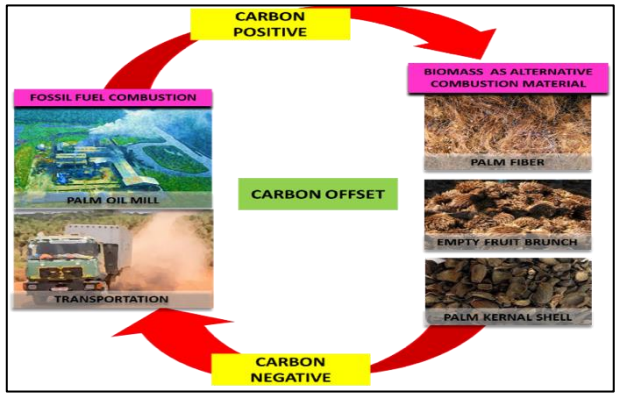
- A carbon footprint refers to the total amount of greenhouse gases (GHG) emitted directly or indirectly by an individual, activity, product, or organization over a specific period of time.
- It is commonly expressed in equivalent tons of carbon dioxide (CO_2-eq).
- In the palm oil industry, carbon footprint is assessed across the entire supply chain, including plantation activities, fertilizer use, mill operations, transportation, and waste treatment processes such as palm oil mill effluent (POME).

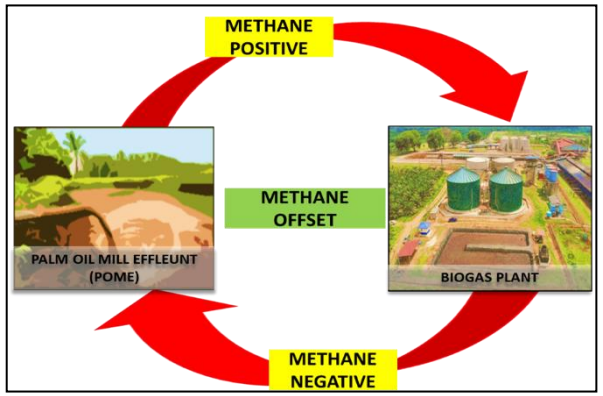
CARBON FOOTPRINT

- Carbon Management in Palm Oil Plantation:

1. Carbon Sequestration (Oil Palm Plantation). Oil palm trees absorb CO_2 from the air during photosynthesis.
2. Biomass as Fuel (PKS, EFB, Fibre). Palm oil waste (PKS, EFB, fibre) is used as fuel to replaces coal and fossil fuels in boilers.
3. Biogas from POME. Biogas plants capture methane from palm oil mill effluent (POME) effluent.







MALAYSIAN ENVIRONMENTAL LAWS QUALITY ACT 1974

- The Environmental Quality Act (EQA) 1974 is a Malaysian law established to prevent, control, and reduce pollution while improving the overall quality of the environment.
- It was introduced because earlier environmental laws (since the 1920s) were limited in scope and unable to manage more complex environmental problems. The Act provides a more comprehensive legal framework for pollution control in Malaysia.
- The Department of Environment (DOE) is responsible for enforcing and administering this Act.

MALAYSIAN ENVIRONMENTAL LAWS QUALITY ACT 1974

- The Act is supported by specific regulations that control:

1. Industrial effluent discharge
2. Sewage discharge
3. Air emissions from industries
4. Motor vehicle emissions
5. Open burning prohibition
6. Scheduled waste management
7. Environmental Impact Assessment (EIA) for major projects

REGULATORY BODIES IN MALAYSIA (ENVIRONMENTAL & WASTE MANAGEMENT)

- Malaysia's environmental and waste management system is controlled by several agencies, each with specific responsibilities to ensure pollution control, waste handling, and sustainable development:

1. **DOE (Department of Environment):** Main enforcement body for pollution control under EQA 1974
2. **Kualiti Alam Sdn. Bhd.:** Licensed contractor for scheduled waste management (collection, treatment, and disposal of hazardous waste)
3. **Jabatan Pengurusan Sisa Pepejal, Kementerian Tenaga dan Sumber Asli:** Responsible for solid waste policy, planning, and national waste management framework.
4. **Local Authorities (PBT):** Handle local waste collection, cleanliness, and enforcement at municipal level

EXAMPLE 3

- What is carbon footprint?
- State 3 regulatory bodies that responsibilities to ensure pollution control and waste handling in Malaysia.

THANK YOU

PLEASE SUBMIT YOUR MIND MAP